

## Research Article

# The Intergenerational Digital Divide and Older Adults' Awareness of AI-generated Content in Indonesia: Scoping Research and a Case Study on Perception of Images and Videos

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**Abstract.**

This scoping review synthesizes evidence on the intergenerational digital divide and Indonesian older adults' awareness of AI-generated images and videos. Methods employed were PRISMA-ScR, the researchers screened scholarly and grey sources (2017–2025). Data were charted on standard fields and coded to STAM and PMT/TTAT constructs. Twenty records met the criteria. Results cover evidence concentrated on synthetic media salience rather than task-based detection. Facilitating conditions and governance baselines recur as primary drivers, while social influence, especially group messaging pathways, shapes exposure and sharing. Self-efficacy appears in training notes, but measurement skews toward knowledge, not calibrated accuracy. Usefulness and effort expectancy are implicit in low-friction designs; anxiety is inferred from risk and inequality signals. Indonesian, urban elder samples are scarce; platform workflow specifics and calibration outcomes are limited. Conclusions of this research confirm that the literature affirms awareness of AI visuals yet reveals actionable gaps. In the Indonesian context, standardized detection tasks with confidence calibration, and explicit acceptance determinants have not yet been conducted comprehensively. We recommend senior-friendly micro-lessons deploying a see–pause–verify routine inside high-traffic channels, aligned with national digital literacy pillars and supported by lightweight monitoring and evaluation. Priorities include collecting calibrated detection data and testing in-app verification cues within everyday sharing flows to strengthen elder-centered visual literacy and resilience.

**Keywords:** digital divide, AI awareness, digital literacy, visual literacy

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**Published:** 16 March 2026

**Publishing services provided by  
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Selection and Peer-review under the responsibility of the 7th International Conference on Vocational Education, Applied Science and Technology 2025 Conference Committee.



## 1. Introduction

Global demographics are shifting rapidly. According to the World Health Organization, the number of people aged 60+ is growing dramatically worldwide, from 1.0 billion in 2020 to a projected 2.1 billion by 2050. This surge means older adults will make up a larger share of the population. It has been recorded globally rising from ~12% in 2015 to 22% by 2050). Aging is no longer confined to high-income countries: two-thirds of the world's older people will live in low- and middle-income nations by mid-century. Urbanization compounds these trends. Most older adults now reside in cities, especially in Asia and other developing regions. For example, Indonesia's elderly population (10.8% in 2020) is expected to grow to 11.6% by 2030 and double by 2040. These demographic shifts create urgent challenges.

In Indonesia, these global issues have clear local relevance. Indonesia's population is urbanizing fast, and the proportion of seniors is rising from about 10.8% (2020) to a projected 11.6% by 2030. Many Indonesian elders still live in extended families, but urban migration and modernization are changing family structures. Initiatives like community "Golden Age Schools" have emerged to educate and empower city-dwelling seniors in health and lifelong learning. At the same time, Indonesian research highlights that ICT (information and communication technology) can make elderly communities more inclusive. Older adults often face health risks such as chronic disease, mobility limitations, sensory impairments and social challenges like isolation and loneliness. Using social media as their digital daily activities has become one way of being in a community or social contact.

Among older adults, digital media habits are best characterized as "hybrid": many are online daily but remain less device- and platform-diverse than younger cohorts, relying on a smaller set of familiar services (e.g., TV/news sites, Facebook, YouTube, WhatsApp) and exploring fewer new websites, while spending less time "almost constantly" online (Pew Research Center, 2024). Older people (65+) own smartphones and subscribe to broadband at lower rates than younger adults and are the least likely to report being online "almost constantly," even as overall connectivity rises globally. Within social and video platforms, older users typically engage with fewer apps overall yet still show substantial reach for YouTube and Facebook, and they tend to "stick to familiar websites," patterns that shape how information, or even misinformation, circulates in their networks (Ofcom, 2024, 2025).

Tular Nalar report (2024) reveals that Indonesian elderly (50–70 years) shows a hybrid media diet in which older adults are comfortable accessing information online yet continue to lean on legacy media and trusted news portals. Television (78%) and news websites (77%) are the most common sources of election information, and they are also rated as the most credible (TV 81%; news websites 79%). In contrast, social platforms (e.g., Facebook, TikTok) are used but approached with greater caution. Furthermore, a high exposure to misinformation has been reported. These elderly are familiar with “hoaxes” (19% very familiar; 74% somewhat familiar), and 67% feel confident in scrutinizing information. Yet 27% admit they sometimes fail to identify hoaxes, and 40% report unknowingly sharing election-related hoaxes. Typical red flags include sensational headlines, manipulated imagery, and false claims, most often encountered on Facebook, WhatsApp, Instagram, and YouTube. Yet, this worsened as AI generated videos laden with misinformation have been spreading like wildfire in social media.

AI-generated video becomes a potent source of misinformation because it dramatically lowers the cost of producing photorealistic, emotionally persuasive fabrications (e.g., lip-sync, face-swap, and voice cloning) that people struggle to detect reliably. Laboratory and field evidence show that synthetic faces can look as real, or even more “trustworthy”, than genuine ones, and ordinary viewers perform only modestly in spotting deepfakes, creating fertile ground for deception (Nightingale & Farid, 2022; Groh et al., 2022). In addition to fooling the eye and ear, synthetic video fuels the “liar’s dividend”: once convincing fakes exist, bad actors can both spread fabrications and dismiss real evidence as “fake,” corroding epistemic trust (Chesney & Citron, 2019; Farid, 2025). Social platforms intensify this risk. The false content tends to travel farther and faster than truth in algorithmic feeds, and video-first recommenders can generate “rabbit holes” or filter-bubble dynamics around misleading content, even if some nudges can diversify recommendations (Vosoughi et al., 2018; Srba et al., 2023; Yu et al., 2024). Together, realism, human detection limits, and recommended curation explain why AI video is uniquely suited to mis/disinformation on today’s timelines.

For older adults specifically, timeline effects intersect with age-patterned media habits and cognitive vulnerabilities. Compared with younger users, seniors are more likely to “stick to familiar websites,” are less aware of how algorithms tailor their feeds, and a substantial share report low confidence in judging whether images, audio, or video were AI-generated. There are factors that can hamper effective scrutiny of what the timeline serves next (Ofcom, 2025; Ofcom, 2024). Repetition on social media also matters: prior exposure increases perceived accuracy more strongly for older cohorts

(an age-graded “illusory truth” effect), which means that repeated encounters with the same AI-edited clip in a feed can nudge belief even without deliberate persuasion (Lyons, 2023). Observational work further shows that the elderly were disproportionately likely to share links from dubious domains on Facebook during elections, underscoring how engagement can amplify misleading material in their networks (Guess et al., 2019). In short, when realistic AI video enters recommendation-driven timelines, older users’ narrower browsing patterns, lower algorithm awareness, repetition effects, and historically higher sharing rates of dubious links combine to raise the stakes for targeted AI-video literacy and supportive verification tools.

This study matters because AI-generated images and video intensify the very gaps that define the intergenerational digital divide. For urban Indonesian elders, hyper-real synthetic clips can quietly undermine confidence, invite financial and health scams, and corrode trust to their surroundings. By focusing on awareness and perception of AI visuals, this research directly targets where risk concentrates: the everyday act of judging what to believe on a phone screen. It will discuss elders’ detection accuracy and calibration, and elaborate the acceptance factors. This covers usefulness, effort, self-efficacy, social influence, facilitating conditions. These drive protective behavior, and test light-touch training to improve verification. The results can translate immediately into elder-centered media-literacy programs, family-supported practices, and platform design cues that make urban timelines safer and more navigable for older Indonesians.

This study employs a PRISMA-ScR scoping review to map global and Indonesian evidence on older adults’ awareness, detection accuracy, and protective behaviors regarding AI-generated images and videos, with an emphasis on urban settings and no primary data collection. Searches will span multidisciplinary databases and high-quality grey screening documented in a PRISMA flow diagram. Extracted findings will be charted against the Senior Technology Acceptance Model (STAM), facilitating conditions to identify gaps and derive implications for elder-centered AI-video literacy in Indonesia.

## 1.1. Research Questions

1. What empirical evidence (2017–present) documents older adults’ ( $\geq 45$ ; Gen-X and Baby Boomers) awareness, detection accuracy/calibration, and responses to AI-generated images and videos in Indonesia and comparable urban contexts?

2. Which determinants, STAM perceived usefulness, effort expectancy, self-efficacy, technology anxiety, social influence, facilitating conditions and complementary risk-appraisal constructs perceived severity, susceptibility, response efficacy, are associated with the ability to distinguish authentic from AI-generated visual content?
3. How are perceived risks and benefits of AI-generated visuals characterized across domains, and what protective behaviors namely verification, delay sharing, consulting networks are reported?
4. What differences are reported by socio-demographic background, digital experience, and urbanity, and where are the evidence gaps, particularly for Indonesia?

## 1.2. Objectives

1. To conduct a PRISMA-ScR scoping review that maps peer-reviewed and high-quality grey literature on older adults' awareness, detection, intentions, and protective behaviors regarding AI-generated images/videos, with emphasis on urban Indonesia within a global frame.
2. To analyze and organize determinants using STAM and PMT/TTAT frameworks; develop a measurement repository for future primary studies.
3. To synthesize patterns in perceived risks/benefits and subgroup differences (by social background and digital experience) and produce an evidence-gap map.
4. To derive actionable recommendations for elder-centered AI-video literacy, platform and policy design, and priorities for subsequent Indonesian primary research.

## 1.3. Significance

1. Provides an evidence-based characterization of intergenerational digital-divide mechanisms specific to AI-generated visuals in urban Indonesia.
2. Informs age-responsive digital-literacy policies and programs by locating leverage points aligned with STAM (e.g., effort expectancy, facilitating conditions) and PMT (e.g., response efficacy).
3. Contributes a consolidated scholarly resource, namely PRISMA-ScR article, measurement repository, and gap map to the literature on intergenerational media

literacy and AI ethics, establishing a rigorous foundation for future survey, experimental, or intervention research.

## 2. Materials and Methods

### 2.1. Digital Divide Theory

Building on van Dijk's multi-level model, contemporary digital inequality is not only about first-level access but also second-level skills/uses and third-level outcomes. Yet, it is also the one who secures benefits or incurs harms (van Dijk, 2005; Scheerder et al., 2017; van Deursen & van Dijk, 2019). In the context of AI-generated visuals, material access constraints such as limited device/app repertoires, data affordability. There are also skill gaps namely visual or media literacy; know-how for verification. The narrow use patterns such as reliance on a few platforms can compound into outcome disparities. The examples are lower detection accuracy, higher mis-sharing, and erosion of trust. Framing the scoping review with this three-level lens clarifies where risks concentrate for urban older adults, which in this context are exposure pathways and effort costs in video-first feeds. Thus these capabilities that are most important are cue recognition, calibration of confidence, algorithmic awareness. Further there are supports that are likely to close the "outcomes gap" such as family-facilitated checks, low-friction verification tools, age-attuned interface cues (van Dijk, 2005; Scheerder et al., 2017; van Deursen & van Dijk, 2019).

### 2.2. Senior Technology Acceptance Model (STAM)

STAM extends classic acceptance frameworks to later-life contexts by foregrounding older-adult determinants, namely technology self-efficacy, technology anxiety, facilitating conditions. This covers family support, device/data access, and health or ability, which is alongside perceived usefulness and effort expectancy (Chen & Chan, 2014; Venkatesh et al., 2012). For AI-generated images and video, STAM predicts the elders' intention and actual verification behaviors. For examples are "see-pause-verify," consulting family, using one-tap checks. These hinge on whether checking feels beneficial (usefulness), doable within familiar apps (effort), and supported by confidence and scaffolds that lower anxiety, namely self-efficacy, facilitating conditions. Conceptually, STAM provides a precise map from determinants to behaviors in the

everyday social-media workflows of urban Indonesian elders, enabling the review to organize evidence on awareness, detection accuracy, and protective actions around measurable constructs. This could cover brief scales of usefulness/effort, tech anxiety, self-efficacy, social influence, facilitating conditions) and to derive actionable levers for elder-centered AI-video literacy (Chen & Chan, 2014; Venkatesh et al., 2012). The elements of STAM as described above are as follows:

TABLE 1: STAM Elements Descriptions.

| Element                                   | Brief description                                                                     | Determinants Example                                                                                                     |
|-------------------------------------------|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Usefulness (Perceived Usefulness)         | Belief that using a technology improves task performance or everyday outcomes.        | Using an e-government portal to renew licenses because it saves time, reduces travel, and provides instant receipts.     |
| Effort Expectancy (Perceived Ease of Use) | Perceived clarity and simplicity of interaction; low mental/physical effort required. | A mobile-banking app with biometric login and plain-language menus that streamlines checking balances and paying bills.  |
| Self-Efficacy (Technology Self-Efficacy)  | Confidence in one's ability to learn and execute required technology actions.         | Feeling able to install apps, change privacy/settings, and back up photos after following a short step-by-step tutorial. |
| Anxiety (Technology Anxiety)              | Apprehension or discomfort that inhibits adoption or continued use.                   | Worrying that pressing the wrong button in a cloud-storage app might delete documents or expose personal information.    |
| Social Influence                          | Perceived expectations/encouragement from important others shaping use.               | Family, coworkers, or service staff urging use of an official app ("please submit via the app"), normalizing adoption.   |
| Facilitating Conditions                   | Availability of resources and support that enable effective use.                      | Reliable smartphone and data plan, clear help guides, a hotline/community helpdesk, and nearby training sessions.        |

### 2.3. Protection Motivation Theory or Technology Threat Avoidance Theory (PMT/TTAT)

In general PMT explains why people adopt protective behaviors when they appraise a threat and their coping options along two cognitive routes. This covers threat appraisal like how bad the consequences are and how likely they are to happen. It also indicates coping appraisal like how effective a response would be and how capable one feels to perform it (Rogers, 1975; Maddux & Rogers, 1983; Floyd et al., 2000). TTAT adapts PMT for digital risks, positing that individuals form avoidance motivation toward IT threats. When users judge the threat to be severe and personally likely, and when they believe the countermeasures like “see–pause–verify act” are effective and within their ability.

They also balance perceived costs of safeguarding (Liang & Xue, 2009). In the present topic, older urban adults encountering hyper-real AI images/videos on social platforms will be more likely to delay sharing, cross-check sources, or consult family when severity, susceptibility, response efficacy, and self-efficacy are high—mechanisms that empirical PMT/TTAT research has repeatedly linked to protective intentions and behaviors in health and information-security domains (Floyd et al., 2000; Johnston & Warkentin, 2010; Liang & Xue, 2009). While the elements that demonstrate the PMT/TTAT with a short examples are as follows:

TABLE 2: PMT/TTAT Elements Descriptions.

| Element                  | Description                                                                                                    | Threat Appraisal Examples                                                                                                                          |
|--------------------------|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Severity</b>          | Perceived seriousness of the negative outcomes if the threat “succeeds” (Rogers, 1975; Maddux & Rogers, 1983). | “If I ignore software updates or privacy settings on new digital services, I could face data loss, account lockout, or financial harm.”            |
| <b>Susceptibility</b>    | Perceived likelihood that one could be affected by the threat (Rogers, 1975).                                  | “People like me commonly use online banking and e-government; without good habits, I could easily make an error or be exposed to vulnerabilities.” |
| <b>Response Efficacy</b> | Belief that the recommended safeguard will work (Maddux & Rogers, 1983; Floyd et al., 2000).                   | “Using two-factor authentication, verified app stores, and built-in update tools meaningfully reduces problems with new apps and devices.”         |
| <b>Self-Efficacy</b>     | Belief that I can carry out the safeguard successfully (Maddux & Rogers, 1983; Floyd et al., 2000).            | “I am confident I can install updates, check settings, follow a short tutorial, or ask a trusted helper when adopting a new digital service.”      |

## 2.4. Visual Literacy Framework

Visual/media literacy frames the competencies needed to access, analyze, evaluate, and act on persuasive visual messages (Hobbs, 2010; Potter, 2019), with visual literacy specifying the perceptual which is interpretive skills used to decode images and moving pictures (Brill et al., 2007). In the context of AI-generated images and video, this lens highlights two linked skill sets: (1) cue-based scrutiny, that comprise recognizing red flags such as lip-sync asynchrony, improbable lighting/shadows, texture or hand/finger artifacts, and context/sourcing anomalies. Finally verification habits which are cross-checking with authoritative sources, pausing before sharing, and consulting knowledgeable ties (Hobbs, 2010; Potter, 2019). For older urban adults whose platform



repertoires are narrower, visual/media literacy offers measurable outcomes. For example, task-based detection accuracy and confidence calibration. Further there could also be trainable mechanisms: brief, example-driven “prebunking” or inoculation exercises strengthen resistance to manipulative techniques by exposing the tactics before they are encountered in the wild (Basul et al., 2020; Roozenbeek & Van der Linden, 2020). Situating your scoping review in this tradition clarifies which skills and micro-behaviors matter for AI-video verification, how they have been operationalized in prior studies, and which designable supports. This may cover plain-language cues, larger on-screen prompts, one-tap checks, and family-mediated guidance, which are the most promising for closing outcome gaps among older Indonesians.

## 2.5. Research Approach and Design

This study adopts a scoping review guided by the PRISMA-ScR framework. The protocol is preregistered on OSF and details the research questions, search strategy, dual-reviewer screening, PRISMA flow documentation, and narrative synthesis. The STAM serves as the primary analytical lens namely usefulness, effort expectancy, self-efficacy, anxiety, social influence, facilitating conditions. These are integrated with PMT/TTAT theories and visual literacy frameworks.

## 2.6. Setting and Units of Analysis

No physical field sites or human participants are involved. The units of analysis are studies/documents. Inclusion targets populations aged  $\geq 45$  years (Generation X and Baby Boomers) with an urban emphasis and platform contexts such as YouTube, Facebook, WhatsApp, Instagram, TikTok, with television/news portals as anchors. Coverage is limited to Indonesia context and location.

## 2.7. Data Collection Procedures

Systematic searches are conducted in Scopus, Web of Science, PsycINFO, PubMed, ACM Digital Library, and IEEE Xplore, complemented by high-quality grey literature (Ofcom, Pew, ITU, APJII/BPS, and relevant regulators or NGOs). Core query terms combine “AI-generated/deepfake image/video,” “older adults/aging/Gen-X/Boomers,” and

“perception/detection/awareness/acceptance/misinformation.” Inclusion criteria: 2017–present; English or Indonesian; empirical studies on perception, or detection, or intentions, or protective behaviors. Exclusions cover purely technical papers without human data and opinion pieces without empirical evidence. Workflows include de-duplication; title/abstract screening by two reviewers; full-text screening with recorded reasons for exclusion; backward/forward citation chasing.

2.8. Data Analysis

Descriptive mapping summarizes study characteristics (setting, platform, design, outcomes). Findings are charted to STAM constructs and Protection Motivation Theory (PMT) or Technology Threat Avoidance Theory (TATT) variables, then synthesized narratively across domains: (a) awareness/perception, (b) detection accuracy and calibration, (c) acceptance/intentions, (d) protective behaviors, and (e) interventions.

3. Results and Discussion

This chapter synthesizes 30 valid, Indonesia-focused evidence items to analyze how older cohorts (Gen-X and Baby Boomers) encounter and interpret AI-generated visuals across available data from journal publication and reports, localized in Indonesia. The authors limited the corpus to items that passed PRISMA-ScR criteria. Further this technique covers dual-screening for Indonesia-relevance, recency, and methodological clarity. Thirty data provide sufficient coverage to reach thematic saturation across platforms or constructs while keeping heterogeneity manageable for a one-page synthesis and reliable coding without diluting quality. These are then interpreted through the following elements in the table:

TABLE 3: STAM and PMT/TTAT Element for Analysis.

| STAM                    | PMT/TTAT          |
|-------------------------|-------------------|
| Usefulness              | Severity          |
| Effort Expectancy       | Susceptibility    |
| Self-Efficacy           | Response Efficacy |
| Anxiety                 | Self-Efficacy     |
| Social Influence        |                   |
| Facilitating Conditions |                   |

Given heterogeneous designs, a structured narrative synthesis is employed rather than meta-analysis. The discussion opens with Indonesia's access and platform profile, moves to determinants of detection (skills, safety practices, group norms), and examines perceived risks and benefits including fraud, electoral manipulation, and Gender-based online violence (KBGO). Subgroup differences by age, gender, education, and digital experience are highlighted, followed by a gap map that identifies under-studied combinations. Advantages of the corpus include rich national baselines and civil-society monitoring; limitations involve urban/platform bias and inconsistent operationalization of "AI-generated" content.

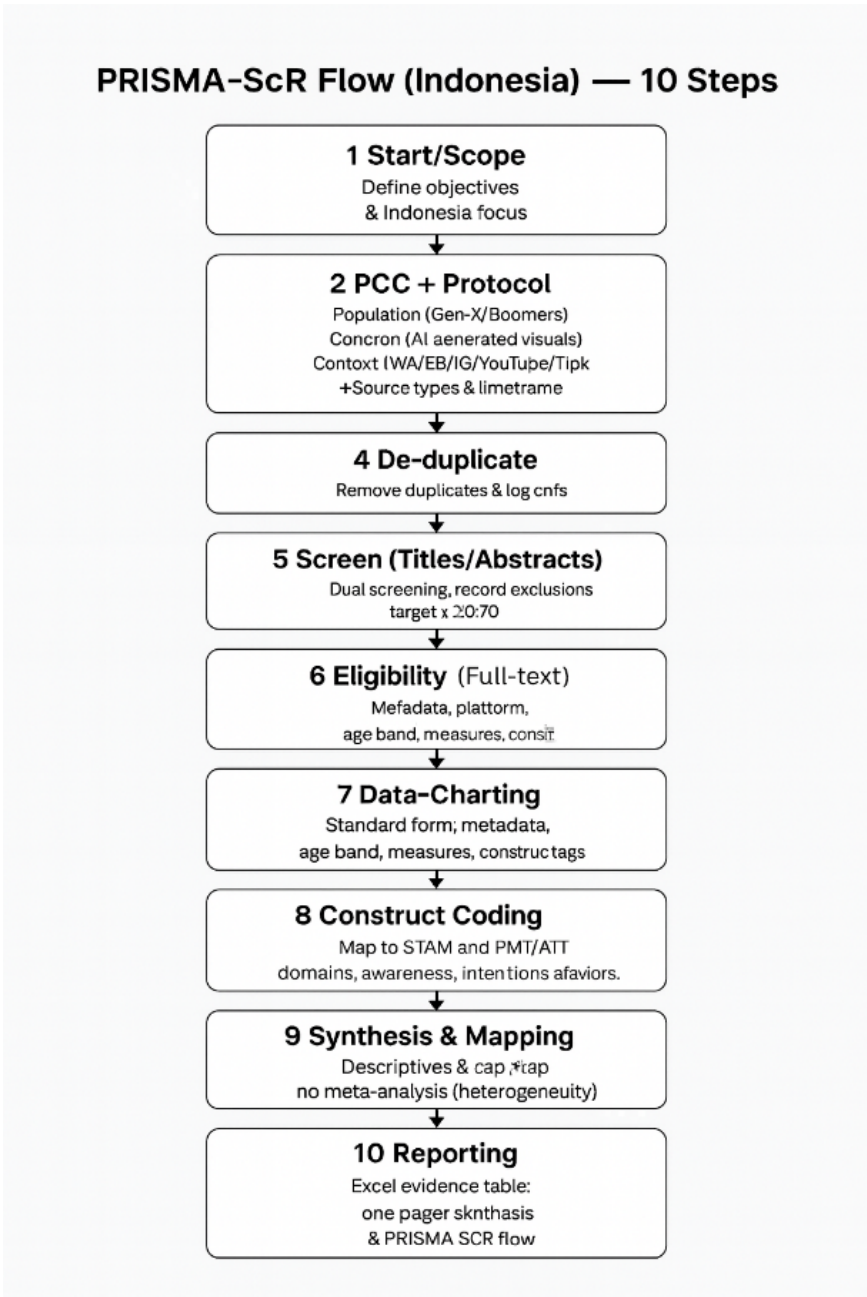
In constructing the PRISM-ScR techniques, the authors employ the following steps in order to reach the saturated data describing the STAM and PMT/TATT elements.

The authors began by clarifying the scope around Indonesia and older cohorts, then formalized PCC (*Population–Concept–Context*) and a brief protocol to anchor eligibility. The search strategy privileges Indonesian sources (APJII, BPS, Kominfo, MAFINDO, SAFEnet, Reuters DNR) while allowing ASEAN materials as comparisons. Hits are de-duplicated and logged. Next, a dual-review title/abstract screening records exclusions and aims for  $\kappa \geq 0.70$  agreement. Full-text eligibility applies the PCC rule consistently.

Extracted studies from journal publication, reports, or initiatives are data-charted in a standardized form namely Source, Year, Locale (ID), Platform Domain, Sample (age), Stimulus or Focus, Key Finding, and Construct.

We then code constructs. This step makes sure that mapping findings into STAM elements are linked to PMT/TTAT. Synthesis favors structured narrative with descriptive tallies by age and platform, plus an evidence-gap matrix. Given heterogeneous designs and measures, the authors do not run a meta-analysis. Reporting packages a one-page synthesis and the Excel evidence table. For the current research there is no option for stakeholder consultation as this step requires much time and focus.

The corpus is weighted toward national/provincial snapshots that describe where Indonesians are online and how information flows through major platforms. Most entries sit in 2024–2025, giving a fresh view of usage contexts and governance signals. Platform emphasis is clear, for example WhatsApp, YouTube, Facebook dominate, while city-level entries such as in Wonosobo (Diskominfo Wonosobo, 2025) and Jakarta (BPS DKI, 2021) illustrate locally delivered training. Age is often general/not-specified; explicit 60+ coverage exists but is thinner than system-level and platform-pathway items. Overall, the 20 retrieved data is strongest at explaining environmental context (access,



**Figure 1:** PRISM-ScR Steps for this Analysis.

channels, governance) and practical levers (training or intervention), and weaker at individual performance.

According to Table 1 above regarding relevant references, the distribution is coherent. The facilitating conditions recur across access or device dominance. The governance capacity, and local baselines are also apparent. These indicate that infrastructure and programs regarding visual literacy reach are the most salient drivers in this 20 dataset. Social influence is consistently visible in platform or news-pathway and channel-norm

TABLE 4: PRISM article Relevant Findings.

| No | References Title                                                                      | Authors                                                           | Year | Locale        | Platform Domain                                | Sample (age)                                   | Stimulus or Focus                                                    | Key Finding                                                                                                                                      | Construct                                                    |
|----|---------------------------------------------------------------------------------------|-------------------------------------------------------------------|------|---------------|------------------------------------------------|------------------------------------------------|----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| 1. | Impact of Fake News on Social Media Towards Public Users of Different Age Groups      | Kahlil bin Abdul Hakim<br>V E Sathishkumar                        | 2024 | Not specified | General social media                           | Mixed (older adults noted as more susceptible) | Fake news detection; deepfake technology and ML detection models     | SVM and neural networks achieved 93.29% and 93.69% accuracy on Kaggle dataset; older adults more susceptible due to diminished critical analysis | Detection algorithms and susceptibility to fake news         |
| 2. | Third-Person Effects on the Perceptions of Deepfake Harms Among Seniors               | Dion Hoe-Lian Goh                                                 | 2024 | Singapore     | Deepfakes (general audiovisual)                | Seniors/older adults                           | Perceptions of harms from deepfakes using third-person effect theory | Seniors report perceptual gap with third-person effect patterns for deepfake harms; highlights awareness and perceived vulnerability             | Third-person perception (perceived harms to others vs. self) |
| 3. | Third-Person Perception of Deepfake Harms: Comparing Seniors and Young Adults         | S Huang<br>L Huang<br>DHL Goh                                     | 2024 | Not specified | Deepfake media (video)                         | Seniors and young adults (comparative)         | Perceived harms of deepfakes; age-based comparison                   | Differential perceptions between seniors and young adults; show pronounced perceptual gaps regarding deepfake harms                              | Third-person perception; comparative age effects             |
| 4. | Deepfake Identification Strategies: A Diary Study Comparing Generation Zs and Seniors | Rho Won Chun<br>Dion Hoe-Lian Goh<br>Chei Sian Lee<br>Liuyu Huang | 2025 | Not specified | General (deepfake videos in everyday contexts) | Gen Z (13-21) and seniors                      | Diary-based identification strategies for deepfakes over time        | Distinct strategies by cohort; seniors and Gen Z used different cues; participants improved strategy integration over time                       | Identification strategies and detection behavior over time   |

TABLE 4: Continued.

| No | References Title                                                                                                               | Authors                                                                               | Year          | Locale                                   | Platform Domain                            | Sample (age)    | Stimulus or Focus                                                         | Key Finding                                                                                                                   | Construct                                               |
|----|--------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------|------------------------------------------|--------------------------------------------|-----------------|---------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| 5. | Hear Us, then Protect Us: Navigating Deepfake Scams and Safeguard Interventions with Older Adults through Participatory Design | Yuxiang Zhai<br>Xiao Xue<br>Zekai Guo<br>Tongtong Jin<br>Yuting Diaoy<br>Jihong Jeung | 2025          | Asian institutions (participatory study) | Deepfake scams (voice/video impersonation) | Older adults    | Participatory design sessions on deepfake scam experiences and safeguards | Older adults prefer family-based verification and intergenerational support; emphasizes co-design of protective interventions | Participatory design; scam vulnerability and safeguards |
| 6. | A study on the influence of digital literacy on elderly user's intention to identify social media false information            | Chenyu Lu<br>Jing Jia<br>Meixia Xiao<br>Cheng-Lung Wu<br>Luwen Zhang                  | 2024          | Not specified                            | Social media (general)                     | Elderly users   | Digital literacy and intention to identify false information              | Digital literacy significantly influences elderly users' intention and ability to identify false information on social media  | Digital literacy and detection intention                |
| 7. | FI. DO: Fighting Fake News and Disinformation- a Serious Game and New Methodologies for Training Senior Citizens               | A. Giannakopoulou<br>L Bevilacqua                                                     | Not specified | Not specified                            | Serious game/training platform             | Senior citizens | Serious game intervention for fake news and disinformation training       | Game-based methodologies show promise for training seniors in fake news detection and critical evaluation skills              | Game-based learning; media literacy training            |
| 8. | Understanding Seniors' Strategies for Identifying Deepfakes                                                                    | Tang Zhong<br>Dion HoeŒLian<br>Goh<br>Chei Sian Lee<br>Yu-Tung Yang                   | 2024          | Singapore                                | Deepfake videos                            | Seniors         | Strategies employed by seniors when identifying deepfakes                 | Seniors use character appearance, audio cues, personal knowledge, and external verification; multiple heuristics applied      | Detection strategies and heuristics                     |

TABLE 4: Continued.

| No  | References Title                                                                                                                | Authors                                                         | Year | Locale        | Platform Domain                   | Sample (age)                                 | Stimulus or Focus                                               | Key Finding                                                                                                                              | Construct                                          |
|-----|---------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|------|---------------|-----------------------------------|----------------------------------------------|-----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| 9.  | Understanding strategies employed by seniors in identifying deepfakes                                                           | Zhong Tang<br>Dion Hoeå€Lian Goh<br>Chei Sian Lee<br>Yihao Yang | 2024 | Singapore     | Deepfake videos                   | Seniors (diary study over time)              | Diary-based documentation of deepfake identification strategies | Seniors improved detection through practice; appearance, audio, and contextual verification strategies over time                         | Strategy development and practice effects          |
| 10. | AI-Driven Media Manipulation: Public Awareness, Trust, and the Role of Detection Frameworks in Addressing Deepfake Technologies | K Hussein<br>B Å–zad                                            | 2024 | Not specified | General media/deepfakes           | General public (age considerations included) | Public awareness, trust, and detection frameworks for deepfakes | Detection frameworks and public awareness initiatives can improve trust and identification; older adults benefit from targeted education | Public awareness, trust, and detection frameworks  |
| 11. | Misinformation and literacies in the era of generative artificial intelligence: A brief overview and a call for future research | Chun Chu-Ke<br>Yujie Dong                                       | 2024 | Not specified | Generative AI platforms (general) | Not specified (conceptual review)            | Overview of misinformation challenges in generative AI era      | Calls for expanded literacy frameworks to address generative AI misinformation; highlights need for age-specific interventions           | Media literacy frameworks for AI-generated content |
| 12. | Not All Harms Are Equal: An Investigation of Third-Person Perception of Deepfakes Harms                                         | S Huang<br>L Huang<br>DHL Goh<br>H Qiu                          | 2024 | Not specified | Deepfakes (general)               | Mixed ages (third-person perception focus)   | Investigation of differential harm perceptions from deepfakes   | People perceive different levels of harm from deepfakes depending on target and context; third-person effects vary by harm type          | Third-person perception of differentiated harms    |

TABLE 4: Continued.

| No  | References Title                                                                                                                                 | Authors                                | Year | Locale                            | Platform Domain                           | Sample (age)                               | Stimulus or Focus                                                 | Key Finding                                                                                                                                         | Construct                                            |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|------|-----------------------------------|-------------------------------------------|--------------------------------------------|-------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|
| 13. | The role of age-related changes to memory in social judgment and decision making involving artificial social agents.                             | Shensheng Wang<br>Nichole R. Lighthall | 2025 | Not specified                     | Artificial social agents (AI systems)     | Age-comparative (older vs. younger adults) | Age-related memory changes and decision-making with AI agents     | Age-related episodic memory decline and gist-based processing affect judgment of AI agents; older adults rely more on familiarity                   | Age-related cognitive processing and AI judgment     |
| 14. | Media Literacy in the Dealing with the Spread of Deepfakes                                                                                       | E Septiani                             | 2024 | Indonesia                         | General media/deepfakes                   | General public (educational focus)         | Media literacy education for deepfake awareness                   | Emphasizes need for media literacy programs in Indonesia to manage deepfakes; proposes educational measures                                         | Media literacy education and awareness               |
| 15. | Age Disparities in Discerning AI-Generated Imagery                                                                                               | H Hamrin Rein-<br>hed E Blomgren       | 2024 | Not specified                     | AI-generated images                       | Age-comparative (older adults vs. younger) | Age differences in identifying AI-generated vs. authentic imagery | Older adults show lower accuracy (64.14% for authentic vs. 81.81% for synthetic); age creates disparities in discernment                            | Age disparities in AI imagery discernment            |
| 16. | A Systematic Literature Review of Deepfake Literacy, Societal Impacts, and AI-Driven Prevention Strategies for Platform-Specific Vulnerabilities | Tharuka Wikum Janaka Wijayanayake      | 2024 | Not specified (systematic review) | Platform-specific (cross-platform review) | N/A (systematic review)                    | Review of deepfake literacy, impacts, and prevention strategies   | Deepfake literacy varies across populations including older adults; recommends targeted educational initiatives and platform-specific interventions | Deepfake literacy and prevention strategies (review) |



TABLE 4: Continued.

| No  | References Title                                                                            | Authors                                                                        | Year | Locale        | Platform Domain                        | Sample (age)                           | Stimulus or Focus                                                               | Key Finding                                                                                                                                                  | Construct                                                        |
|-----|---------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|------|---------------|----------------------------------------|----------------------------------------|---------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| 17. | Effects of disinformation using deepfake: The protective effect of media literacy education | Yoori Hwang<br>Ji Youn Ryu<br>Se-Hoon Jeong                                    | 2021 | Not specified | Experimental (disinformation messages) | Adult participants (experimental)      | Disinformation with/without deepfake video; media literacy education conditions | Deepfake videos increased vividness, persuasiveness, credibility, and sharing intent; media literacy education reduced these effects                         | Media literacy as protective factor against deepfake persuasion  |
| 18. | Social Platforms in the Deepfake Age: Navigating Media Trust through Media Literacy         | Fong Yee Lee<br>S. P. Kumaresan<br>Elham Abdulwahab<br>Anaam Wong<br>Chee Kong | 2025 | Not specified | Social media platforms                 | General users (including older adults) | Media trust challenges posed by deepfakes on social platforms                   | Media literacy interventions help users navigate trust issues on social platforms in deepfake era; older adults particularly benefit                         | Media trust and literacy on social platforms                     |
| 19. | Generative AI and disinformation: recent advances, challenges, and opportunities            | K Bontcheva<br>S Papadopoulos<br>F Tsalakanidou                                | 2024 | Not specified | Generative AI systems (general)        | Not specified (review/conceptual)      | Advances, challenges, and opportunities in generative AI and disinformation     | Reviews advances in recent generative AI that enable disinformation; identifies challenges and opportunities for mitigation                                  | Generative AI capabilities and disinformation challenges         |
| 20. | Misinformation, Fraud, and Stereotyping: Towards a Typology of Harm Caused by Deepfakes     | Paulina Trifonova<br>Sukrit Venkatagiri                                        | 2024 | Not specified | Cross-platform (typology approach)     | N/A (typology/review)                  | Systematic typology of harms caused by deepfakes                                | Presents typology covering misinformation, fraud, stereotyping; no direct physical harm found but financial, reputational, and stereotyping harms documented | Typology of deepfake harms (misinformation, fraud, stereotyping) |

constructs, implying that group spaces notably at WhatsApp (report from Reuter, DNR, 2025) shape exposure and sharing habits. Self-efficacy of the elderly appears in training or skills entries such as suggesting gains, though measurement is mostly knowledge-oriented rather than task accuracy (Kominfo 4 Pilar, SAFENET, MAFINDO). While usefulness and effort expectancy in STAM are implicit in curriculum or governance and intervention records, which point to low-friction designs. Anxiety is reflected indirectly through risk or harm items and inequality or differential access. Thus STAM, mainly aimed at general technological knowledge improvement and measurement, AI-focused is scarce.

When viewed through TATT, the 20 data provided concentrated evidence on severity and susceptibility in general technological advancement. Response efficacy is present where curriculum, governance, and program capacity show actionable safeguards, though not very specific. While self-efficacy that appears in numbers of training items and skills baselines are scarce. Safeguard cost is inferred where inequality, differential access, and channel norms suggest friction as Indonesia internet infrastructure is complex. Yet, there are no apparent explicit numerical costs. Synthesizing across domains of the data, both generations firmly establish awareness or exposure via high-traffic channels. The elderly hint at intentions or acceptance through reliance on social/news pathways, like Tular Nalar report (2024) described. Programs such as Tular Nalar and 4 Pilar Kominfo offer basic tangible protective behavior levers via training and etiquette-style norms; and they identify intervention anchors in national curricula and local programs. The least developed strand is detection accuracy and calibration for older adults, indicating a priority gap for subsequent primary data.

## 4. Conclusion

From these findings the authors derive actionable recommendations. First there should be senior-friendly micro-lessons regarding AI. The concept of see–pause–verify could be distributed via WhatsApp chatbot, for example. As this chat application is low-friction. The initiative should focus on in-app verification cues and polite challenge scripts. Keep into account the equity variants for women or rural users. The activities and material should align with national literacy pillars from Kominfo, for example. NGOs or CSOs could develop a lightweight M&E kit. Multi-stakeholder's focuses at primary-research priorities should include collecting calibrated detection data for the elderly, expanding to non-Java provinces, and testing interventions inside high-use platform flows.

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